< CSS3 >

Cascading Style Sheets Level 3

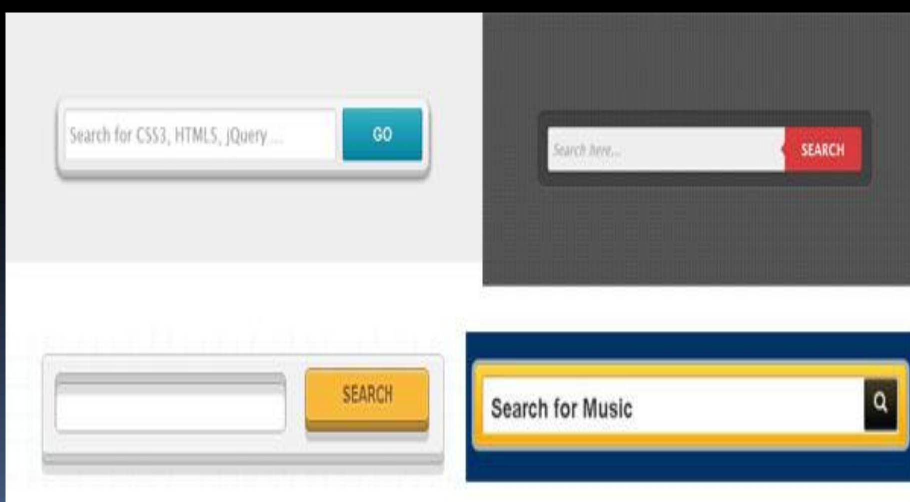
CSS3 Introduction

- CSS3 is the latest standard for CSS.
- **CSS**, or **Cascading Styles Sheets**, is a way to style and present HTML.
- HTML is the **meaning** or **content**, the style sheet is the **presentation** of that document.
- Styles don't smell or taste anything like HTML, they have a format of '**property: value'** and most properties can be applied to most HTML tags.

CSS3


CSS3
Navigation Menu


BORDER-RADIUS
CSS3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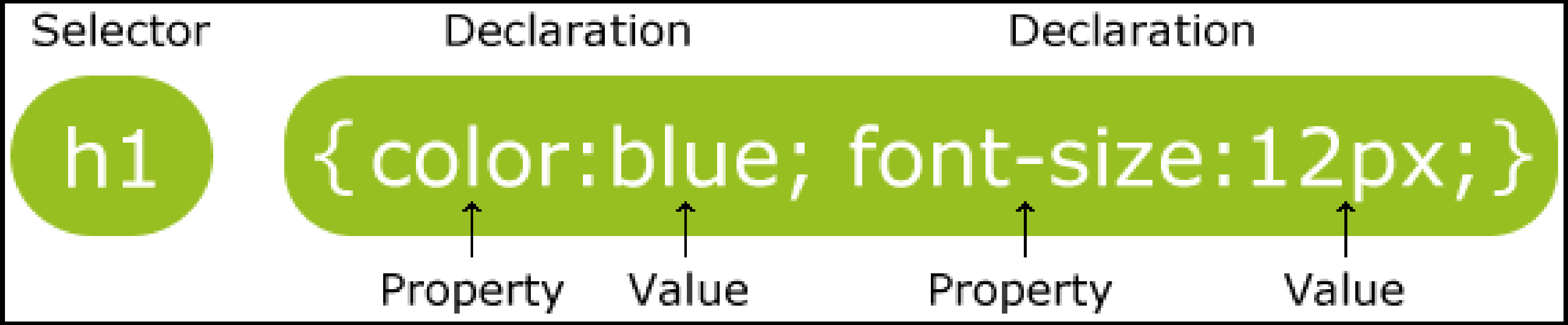
CSS3 REFLECTION

CSS3 modules

- Selectors
- Box Model
- Backgrounds and Borders
- Image Values and Replaced Content
- Text Effects
- 2D/3D Transformations
- Animations
- Multiple Column Layout
- User Interface

CSS3 Syntax

- A CSS rule set consists of a selector and a declaration block.



•The selector points to the HTML element you want to style.

•The declaration block contains one or more declarations separated by semicolons.

•Each declaration includes a property name and a value, separated by a colon.

Selectors

- CSS selectors allow you to select and manipulate HTML element(s).
- CSS selectors are used to "find" (or select) HTML elements based on their id, classes, types, attributes, values of attributes and much more.

- Element Selectors
- Class Selectors
- ID Selectors

CSS3 Backgrounds

- CSS3 contains several new background properties.
- It allows greater control of the background element.
- **background-size**
- **background-origin**
- **Use multiple background images.**

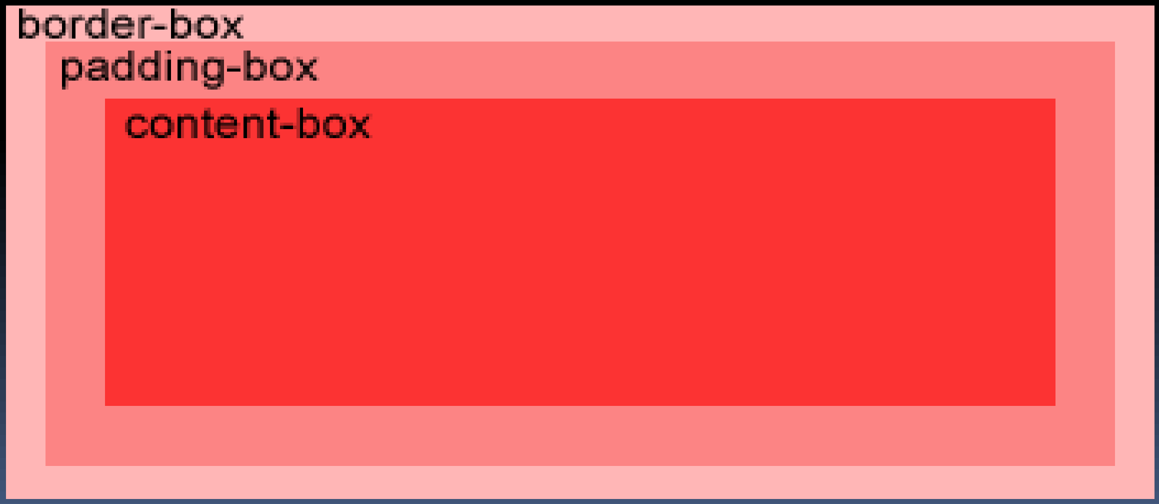
background-size Property

- The background-size property specifies the size of the background image.

```
div {
    background: url(img_flwr.gif);
    background-size: 80px 60px;
    background-repeat: no-repeat;
}
```

background-origin Property

- The background-origin property specifies the positioning area of the background images.
- The background image can be placed within the content-box, padding-box, or border-box area.



Background image within the content-box

```
div {
    background: url(img_flwr.gif);
    background-repeat: no-repeat;
    background-size: 100% 100%;
    background-origin: content-box;
}
```

Multiple Background Images

```
body {
   background: url(img_tree.gif),
     url(img_flwr.gif);
   background-size: 100% 100%;
   background-repeat: no-repeat;
}
```

Text Effects

- CSS3 contains several new text features.
- text-shadow
- word-wrap

Text shadow effect!

- We specify the horizontal shadow, the vertical shadow, the blur distance, and the color of the shadow.

```
h1 {
   text-shadow: 5px 5px 5px #FF0000;
}
```

Word Wrapping

- If a word is too long to fit within an area, it expands outside.
- The word-wrap property allows you to force the text to wrap - even if it means splitting it in the middle of a word

```
p {
  word-wrap: break-word;
}
```

This paragraph contains a very long word: thisisaveryveryveryveryvery verylongword. The long word will break and wrap to the next line.

CSS3 Text Properties

Property	Description
hanging-punctuation	Specifies whether a punctuation character may be placed outside the line box
punctuation-trim	Specifies whether a punctuation character should be trimmed
text-align-last	Describes how the last line of a block or a line right before a forced line break is aligned when text-align is "justify"
text-emphasis	Applies emphasis marks, and the foreground color of the emphasis marks, to the element's text
text-justify	Specifies the justification method used when text-align is "justify"
text-outline	Specifies a text outline
text-overflow	Specifies what should happen when text overflows the containing element
text-shadow	Adds shadow to text
text-wrap	Specifies line breaking rules for text
word-break	Specifies line breaking rules for non-CJK scripts
word-wrap	Allows long, unbreakable words to be broken and wrap to the next line

CSS3 Web Fonts

With CSS3, web designers are no longer forced to use only "web-safe" fonts.

- Web fonts allow Web designers to use fonts that are not installed on the user's computer.
- When you have found/bought the font you wish to use, just include the font file on your web server, and it will be automatically downloaded to the user when needed.
- Your "own" fonts are defined within the **CSS3 @font-face rule.**

The @font-face Rule

- We must first define a name for the font (e.g. myFirstFont), and then point to the font file.
- To use the font for an HTML element, refer to the name of the font (myFirstFont) through the font-family property.

```
@font-face {
  font-family: myFirstFont;
  src: url(sansation_light.woff);
}

div {
  font-family: myFirstFont;
}
```

CSS3 Links

- The link element can have different style depending on which state it is.
  - **a:link - not visited link**
  - **a:visited – visited link**
  - **a:hover - a link when the user hover the mouse over it**
  - **a:active - a link when it is pressed**

Example - Links

a:link {color:#1BA1E2;}

a:visited {color:#00FF02;}

a:hover {color:#FF00FF;}

a:active {color:#2200FF;}

Rules

The following rules apply for styling links:

- a:hover must be used
  after a:link and a:visited
- a:active must be used after a:hover

Common Link Styling Properties

The following properties are most widely used for styling of links:

- color
- text-decoration
- background-color

```
a:link{text-decoration:underline;}
a:hover {background-color:#FF00FF;}
a:visited {color:#00FF2D;}
```

Borders

- With CSS3, you can create rounded borders, add shadow to boxes, and use an image as a border - without using a design program, like Photoshop.

- **border-radius**
- **box-shadow**
- **border-image**

Border-radius Property - Rounded Corners

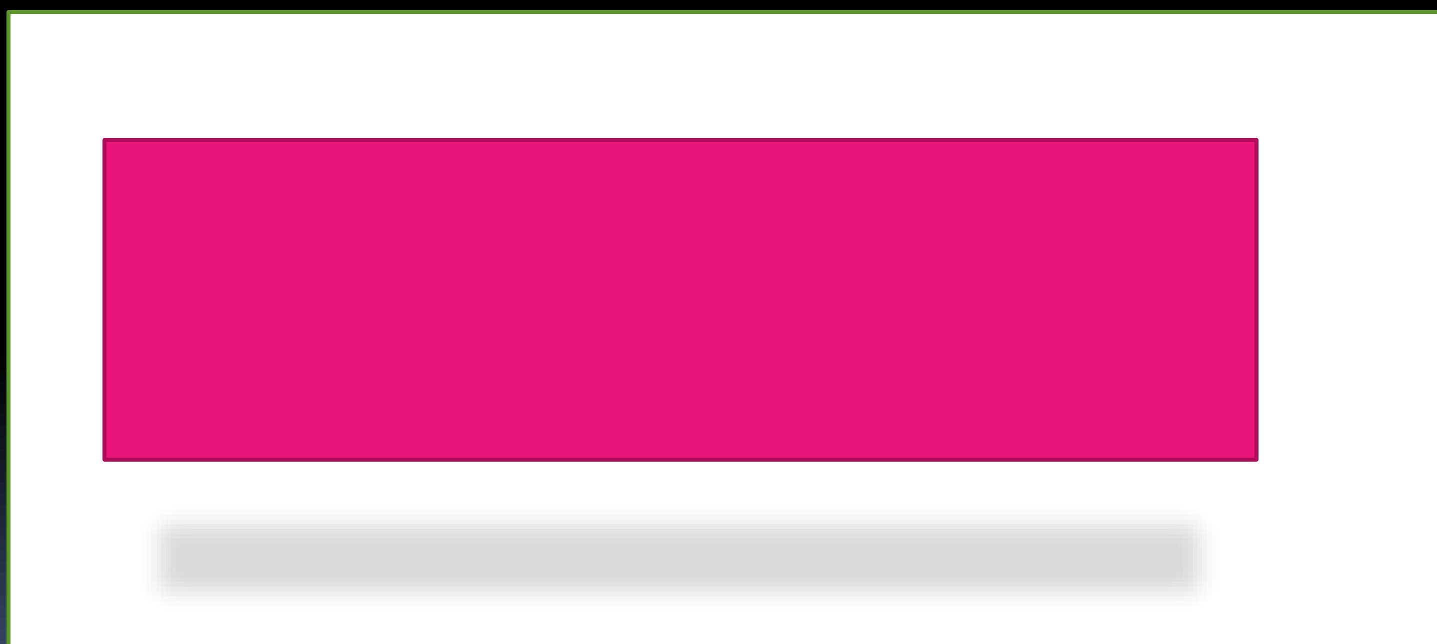
- The border-radius property is used to create rounded corners.

Add rounded corners to a div element:

```
div {
    border: 2px solid;
    border-radius: 25px;
}
```

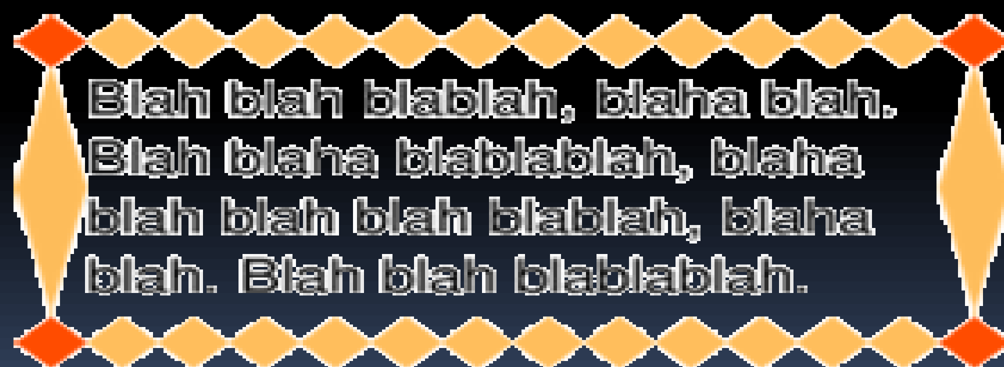
Box-shadow Property

- div {
    box-shadow: 10px 10px 5px #888888;
  }



Border-image Property

- With the CSS3 border-image property you can use an image to create a border.
- The border-image property allows you to specify an image as a border!
- The original image used to create the border below



Use an image to create a border around a div element

```
div {
   -webkit-border-image: url(border.png) 30 30 round; /* Safari 5 */
   -o-border-image: url(border.png) 30 30 round; /* Opera 10.5-12.1 */
   border-image: url(border.png) 30 30 round;
}
```

CSS3 Border Properties

Property	Description
border-image	A shorthand property for setting all the border-image-* properties
border-radius	A shorthand property for setting all the four border-*-radius properties
box-shadow	Attaches one or more drop-shadows to the box

Gradients

- CSS3 gradients let you display smooth transitions between two or more specified colors.
- Earlier, you had to use images for these effects.
- By using CSS3 gradients you can reduce download time and bandwidth usage.
- In addition, elements with gradients look better when zoomed, because the gradient is generated by the browser.

Types of gradients

Linear Gradients

(goes down/up/left/right/diagonally)

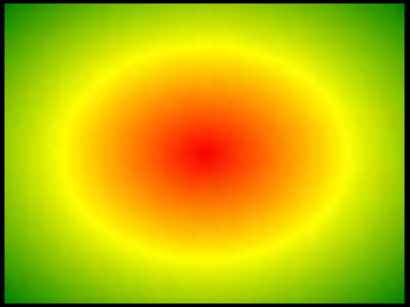
Radial Gradients
(defined by their center)

Linear Gradients

- To create a linear gradient you must define at least two color stops.
- Color stops are the colors you want to render smooth transitions among.
- You can also set a starting point and a direction (or an angle) along with the gradient effect.

background: linear-gradient(*direction*, *color-stop1*, *color-stop2*, ...);

Radial Gradients



- A radial gradient is defined by its center.
- To create a radial gradient you must also define at least two color stops.
- You can also specify the gradient's center, shape (circle or ellipse) as well as its size.
- By default, center is center, shape is ellipse, and size is farthest-corner.

background: radial-gradient(*center, shape size, start-color, ..., last-color*);

CSS3 Transforms

- With CSS3 transform, we can move, scale, turn, spin, and stretch elements.
- A transformation is an effect that lets an element change shape, size and position.
- You can transform your elements using 2D or 3D transformation.

CSS3
Rotate

CSS3
Scale

2D Transforms

- translate()
- rotate()
- scale()
- skew()
- matrix()

Rotate()

- With the rotate() method, the element rotates clockwise at a given degree.
- Negative values are allowed and rotates the element counter-clockwise.

```
div {
    -ms-transform: rotate(30deg); /* IE 9 */
    -webkit-transform: rotate(30deg); /* Chrome, Safari, Opera */
    transform: rotate(30deg);
}
```

- The value rotate(30deg) rotates the element clockwise 30 degrees.

The translate() Method

- With the translate() method, the element moves from its current position, depending on the parameters given for the left (X-axis) and the top (Y-axis) position.

```
div {
    -ms-transform: translate(50px,100px); /* IE 9 */
    -webkit-transform: translate(50px,100px); /* Chrome, Safari, Opera */
    transform: translate(50px,100px);
}
```

- The value translate(50px,100px) moves the element 50 pixels from the left, and 100 pixels from the top.

The scale() Method

- With the scale() method, the element increases or decreases the size, depending on the parameters given for the width (X-axis) and the height (Y-axis).

```
div {
    -ms-transform: scale(2,4); /* IE 9 */
    -webkit-transform: scale(2,4); /* Chrome, Safari, Opera */
    transform: scale(2,4);
}
```

- The value scale(2,4) transforms the width to be twice its original size, and the height 4 times its original size.

The skew() Method

- With the skew() method, the element turns in a given angle, depending on the parameters given for the horizontal (X-axis) and the vertical (Y-axis) lines

```
div {
    -ms-transform: skew(30deg,20deg); /* IE 9 */
    -webkit-transform: skew(30deg,20deg); /* Chrome, Safari, Opera */
    transform: skew(30deg,20deg);
}
```

- The value skew(30deg,20deg) turns the element 30 degrees around the X-axis, and 20 degrees around the Y-axis.

The matrix() Method

- The matrix() method combines all of the 2D transform methods into one.
- The matrix method take six parameters, containing mathematic functions, which allows you to: rotate, scale, move (translate), and skew elements.

```
div {
    -ms-transform: matrix(0.866,0.5,-0.5,0.866,0,0); /* IE 9 */
    -webkit-transform: matrix(0.866,0.5,-0.5,0.866,0,0); /* Chrome, Safari, Opera */
    transform: matrix(0.866,0.5,-0.5,0.866,0,0);
}
```

2D Transform Methods

Function	Description
matrix(*n,n,n,n,n,n*)	**Defines a 2D transformation, using a matrix of six values**
translate(*x,y*)	**Defines a 2D translation, moving the element along the X- and the Y-axis**
translateX(*n*)	**Defines a 2D translation, moving the element along the X-axis**
translateY(*n*)	**Defines a 2D translation, moving the element along the Y-axis**
scale(*x,y*)	**Defines a 2D scale transformation, changing the elements width and height**
scaleX(*n*)	**Defines a 2D scale transformation, changing the element's width**
scaleY(*n*)	**Defines a 2D scale transformation, changing the element's height**
rotate(*angle*)	**Defines a 2D rotation, the angle is specified in the parameter**
skew(*x-angle,y-angle*)	**Defines a 2D skew transformation along the X- and the Y-axis**
skewX(*angle*)	**Defines a 2D skew transformation along the X-axis**
skewY(*angle*)	**Defines a 2D skew transformation along the Y-axis**

3D Transforms

- CSS3 allows you to format your elements using 3D transforms.

rotateX()

rotateY()

The rotateX() Method

- The element rotates around its X-axis at a given degree.

```
div {
    -webkit-transform: rotateX(120deg); /* Chrome, Safari, Opera */
    transform: rotateX(120deg);
}
```

The rotateY() Method

- With the rotateY() method, the element rotates around its Y-axis at a given degree.

```
div {
    -webkit-transform: rotateY(130deg); /* Chrome, Safari, Opera */
    transform: rotateY(130deg);
}
```

3D Transform Methods

Function	Description
matrix3d (*n,n,n,n,n,n,n,n,n,n,n,n, n,n,n,n*)	**Defines a 3D transformation, using a 4x4 matrix of 16 values**
translate3d(*x,y,z*)	**Defines a 3D translation**
translateX(*x*)	**Defines a 3D translation, using only the value for the X-axis**
translateY(*y*)	**Defines a 3D translation, using only the value for the Y-axis**
translateZ(*z*)	**Defines a 3D translation, using only the value for the Z-axis**
scale3d(*x,y,z*)	**Defines a 3D scale transformation**
scaleX(*x*)	**Defines a 3D scale transformation by giving a value for the X-axis**
scaleY(*y*)	**Defines a 3D scale transformation by giving a value for the Y-axis**
scaleZ(*z*)	**Defines a 3D scale transformation by giving a value for the Z-axis**
rotate3d(*x,y,z,angle*)	**Defines a 3D rotation**
rotateX(*angle*)	**Defines a 3D rotation along the X-axis**
rotateY(*angle*)	**Defines a 3D rotation along the Y-axis**
rotateZ(*angle*)	**Defines a 3D rotation along the Z-axis**
perspective(*n*)	**Defines a perspective view for a 3D transformed element**

CSS3 Transitions

- We can add an effect when changing from one style to another, without using Flash animations or JavaScripts.
- CSS3 transitions are effects that let an element gradually change from one style to another.
- To do this, you must specify two things:
- The CSS property you want to add an effect to
- The duration of the effect.

```
div {
    -webkit-transition: width 2s; /* For Safari 3.1 to 6.0 */
    transition: width 2s;
}
```

Animations

- We can create animations which can replace Flash animations, animated images, and JavaScripts in existing web pages.
- The @keyframes rule is where the animation is created.
- Specify a CSS style inside the @keyframes rule and the animation will gradually change from the current style to the new style.

Animation. . .

- When an animation is created in the @keyframe rule, you must bind it to a selector, otherwise the animation will have no effect.
  - Bind the animation to a selector (element) by specifying at least these two properties:
  - the name of the animation
  - the duration of the animation

```
div {
    -webkit-animation: myfirst 5s; /* Chrome, Safari, Opera */
    animation: myfirst 5s;
}
```

CSS @keyframes Rule

- The @keyframes rule specifies the animation code.
- The animation is created by gradually changing from one set of CSS styles to another.
- During the animation, you can change the set of CSS styles many times.
- Specify when the style change will happen in percent, or with the keywords "from" and "to", which is the same as 0% and 100%. 0% is the beginning of the animation, 100% is when the animation is complete.

@keyframes *animationname {keyframes-selector {css-styles;}}*

CSS3 Multiple Columns

- We can create multiple columns for laying out text - like in newspapers!

- Multiple column properties:
  - column-count
  - column-gap
  - column-rule

Create Multiple Columns

- The column-count property specifies the number of columns an element should be divided into:

Example

- Divide the text in a div element into three columns:

```
div {
    -webkit-column-count: 3; /* Chrome, Safari, Opera */
    -moz-column-count: 3; /* Firefox */
    column-count: 3;
}
```

Multiple Columns Properties

Property	Description
column-count	Specifies the number of columns an element should be divided into
column-fill	Specifies how to fill columns
column-gap	Specifies the gap between the columns
column-rule	A shorthand property for setting all the column-rule-* properties
column-rule-color	Specifies the color of the rule between columns
column-rule-style	Specifies the style of the rule between columns
column-rule-width	Specifies the width of the rule between columns
column-span	Specifies how many columns an element should span across
column-width	Specifies the width of the columns
columns	A shorthand property for setting column-width and column-count

CSS Media Query

- With users browsing the web on screens of all sizes – from mobile phones to widescreen monitors, designing flexible layouts has become a key part of modern web development.
- CSS Media Queries help address this need by allowing developers to apply different styles based on a device’s screen size, orientation, and other features.

What is CSS Media Query?

- CSS Media Query is a CSS technique used to apply styles based on a device's characteristics like screen size, resolution, or orientation.

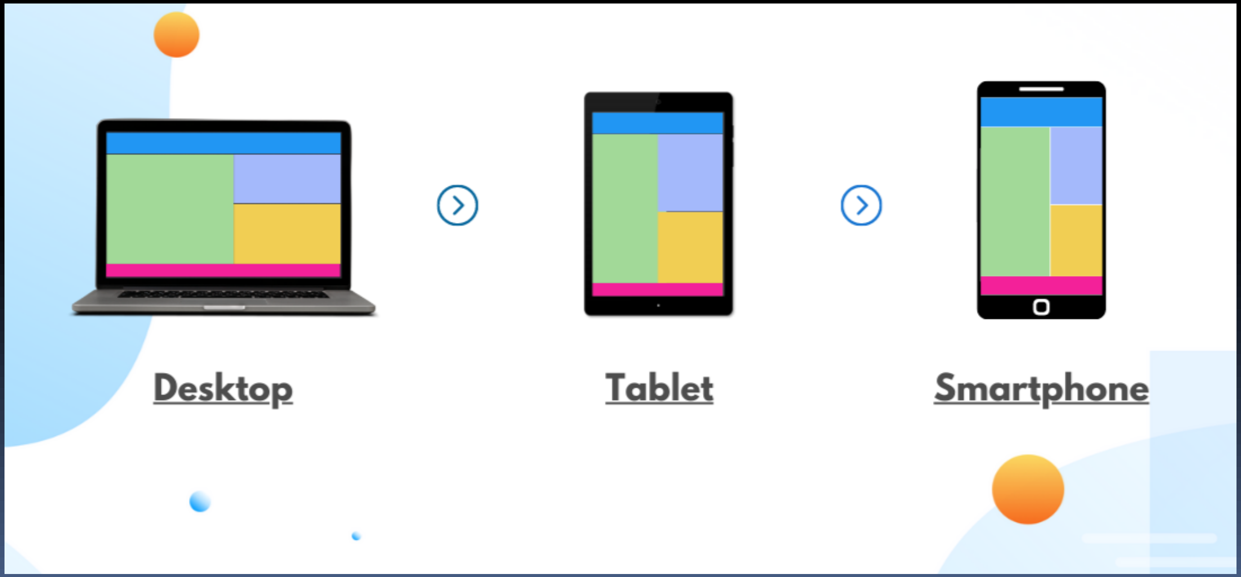
- Why use CSS Media Query?
  - Enhanced User Experience
  - Search Engine Optimization
  - Better Reach of Website

What is CSS Media Query?

- CSS Media query is a technique for modifying a website's behavior and appearance based on the user's device conditions, such as display size, orientation, resolution, the type of device used, and other browser settings.
- Other ways to access Media Queries are **HTML** and **JavaScript**.
- However, CSS offers the most features and is the most popular place to script Media Queries for a website.

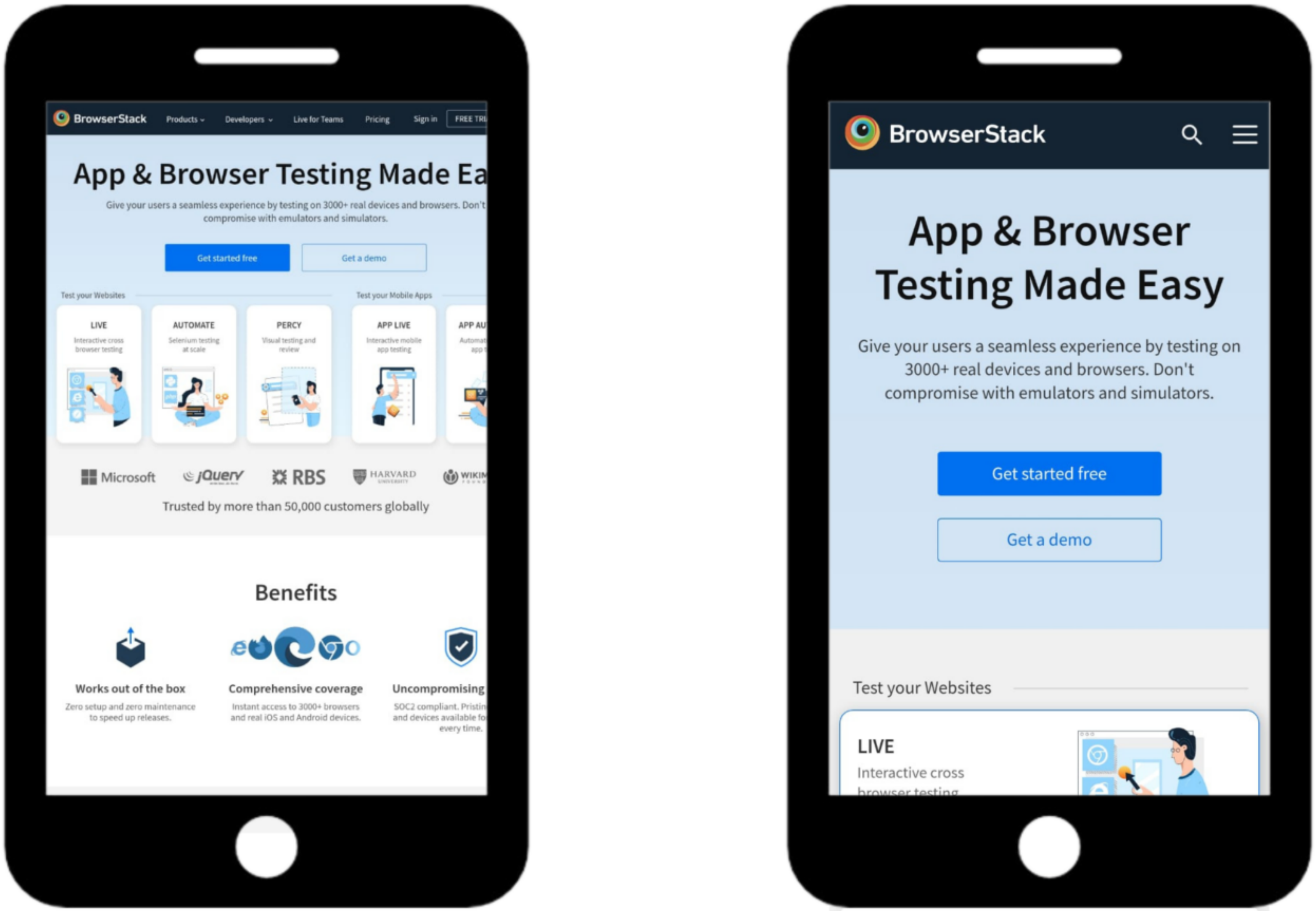
What is CSS Media Query?

- Instead of targeting device types, CSS Media Queries adapt layouts for desktops, laptops, tablets, and phones using the **@media** rule.



Why use CSS Media Query?

- CSS Media Queries are key for building responsive, accessible, and user-friendly websites across all devices.
  - **Better User Experience Across Devices**
  - **Improved Accessibility**
  - **Targeting Specific Screen Sizes Strategically**
  - **Enhanced SEO and Mobile Friendliness**
  - **Wider Audience Reach**
  - **Future-Ready Design**
  - **Content Prioritization**



non responsive vs responsive website

CSS Media Query Syntax

- A media query uses the **@media** rule followed by a media type and one or more conditions.
- If the conditions are true, the styles inside the block are applied.
- In this example, the styles apply only when the screen width is 768px or less.

```css
@media media-type and (condition) {
  /* CSS rules */
}
```

Example:

```css
@media screen and (max-width: 768px) {
  body {
    font-size: 14px;
  }
}
```

END